

ARNAB LAHIRY (M.S., PHYSICS)

DOCTORAL RESEARCHER (PHYSICS | COMPUTER SCIENCE)

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INTRODUCTION

I have **extensive experience bridging scientific research and advanced computation**. Proficient in handling **large, complex datasets**, **data preprocessing**, **statistical modelling**, and **machine and deep learning pipelines** for scientific insight. I design and implement **scalable algorithms**, **numerical models**, and **distributed / high-performance computing solutions**. A proven **collaborative team player**, I combine **rigorous coding practices**, **rapid learning**, and **creative problem-solving** to deliver high-impact results. Leveraging **strong oral and written communication** and **data-visualisation skills** to drive cross-disciplinary projects.

TECHNICAL SKILLS

Programming: Python (Primary), C, C++, FORTRAN, MATLAB

Machine Learning / AI: PyTorch, PyG, Lightning, TensorFlow, Keras, Scikit-Learn, vLLM, Jax, Optuna, Captum

Numerical / Scientific Computing: NumPy, Astropy, SciPy, h5py, Pandas, SymPy, Cython, PyWavelets

Visualization: Matplotlib, Seaborn, Corner, Plotly, PyVista

Collaboration & Tools: Git, GitHub, GitLab, Jupyter, VSCode, L^AT_EX, MS Office, HPC (SLURM), HTML5, CSS3

Cloud & Deployment: SQL, SQLite, MySQL, Docker, Kubernetes, CUDA, AWS (EC2, S3, SageMaker)

Operating Systems: Linux (Ubuntu, Debian), macOS, iOS, iPadOS, Android, Windows

Statistical Techniques: Bayesian inference & optimisation, Gaussian processes, time series analysis, simulation-based inference

EDUCATION

University of Crete

Ph.D. in Physics

Heraklion, Crete, Greece

Nov 2023 – Present

Indian Institute of Science Education and Research (IISER) Tirupati

B.S. - M.S. Dual Degree in Physics

Tirupati, Andhra Pradesh, India

Aug 2018 – Jul 2023

CORE EXPERIENCES AND PROJECTS

Doctoral Researcher

Foundation for Research and Technology - Hellas (FORTH), Crete, GR

Nov 2023 – Present

Institutes of Computer Science & Astrophysics

Visiting Doctoral Researcher at LCS, Département d'Astrophysique, UMR AIM, IRFU,

Commissariat à l'énergie atomique et aux énergies alternatives (CEA) Paris-Saclay, FR

- Built scalable **3D spectral datacube** and **multi-dimensional dataset** pipelines from **theoretical physical models** of galaxies, **FITS** and **HDF5** sources for AI workflows.
- Developed an **iterative 2D–1D wavelet-based algorithm** using **multiresolution feature engineering** for 3D signal denoising.
- Designed and trained a **3D U-Net** with **local convolutions**, **skip connections**, and **anisotropic multiscale learning** to denoise 3D galaxy datacubes, achieving **>95% intensity conservation** and **3–6 σ signal-to-noise improvement**.
- Implemented **fully connected** and **1D CNN architectures** for feature inference from galaxy spectra, evaluated via **RMSE metrics** and cross-model consistency checks.
- Created modular **data ingestion, scaling, and normalization pipelines** ensuring reproducible, high-quality preprocessing.

Masters Research Intern

Flatiron Institute, Simons Foundation, NY, USA

May 2022 – May 2023

Center for Computational Astrophysics (CCA)

- Applied **2D CNNs** and **Graph Neural Networks (GNNs)** for parameter inference on simulated cosmological image datasets.
- Built efficient **data preprocessing pipelines** with **log-scaling**, **Z-score normalization**, and **parameter standardization**.
- Designed and tuned **CNN architectures** with **batch normalization**, **dropout**, **learning rate scheduling**, and **custom log-scaled loss functions**, **hyperparameters optimised** using the Bayesian Tree-structured Parzen Estimator (**TPE**) **Sampler**.
- Implemented **multi-layer message-passing GNNs** on graph-structured data linking galaxies and dark matter haloes.
- Applied **interpretability algorithms** (Saliency, Integrated Gradients, Gradient SHAP) to quantify feature importance.

PUBLICATIONS

- “Interpreting Cosmological Information from Neural Networks in the Hydrodynamic Universe” published at the *Astrophysical Journal* (2025) [DOI:10.3847/1538-4357/ae0b5a] [arXiv:2504.17839v1].
- “Deep and Sparse Denoising Benchmarks for Spectral Data Cubes of High-z Galaxies: From Simulations to ALMA Observations” accepted at *Astronomy & Astrophysics* (2025) [arXiv:2602.10893].

RESEARCH GRANTS

Ph.D. supported by the TITAN ERA Chair project (contract no. 101086741) under the Horizon Europe Programme of the European Commission.

TEACHING, WORKSHOPS, AND COMMUNICATION

Astrostatistics Summer School - Crete <i>Teaching Assistant</i> Assisted in lectures on classical and Bayesian statistics, optimisation methods, MCMC, Gaussian processes, clustering, deep learning, simulation-based inference, GPU parallelisation.	Heraklion, Crete, GR 16–20 June 2025
CODE / ASTRO Software Development Workshop <i>Participant Contributor</i> - Learned GitFlow, astronomical software development, Python package building, PyPI deployment, and documentation - Contributed group project on developing an end-to-end software package, version control in GitHub, deployment and hosting on PyPi [GitHub Repository]	Evanston, IL, USA 4–8 August 2025
TITAN Astroinformatics Group, FORTH <i>Member Contributor</i> Presented research in multidisciplinary collaboration meetings	Heraklion, GR Paris, FR
Relevant Conferences <i>Participant Contributor</i> - UniverseAI: Exploring the Universe with Artificial Intelligence - ML4ASTRO2: Machine Learning for Astrophysics – 2nd Edition - COSMO21: Statistical Challenges in 21st Century Cosmology	[2–6 June 2025] Athens, GR [8–12 July 2024] Catania, IT [20–24 May 2024] Chania, GR

PERSONAL INFORMATION

Nationality: Indian **Date of Birth:** 28 March 2001 **Languages:** English (Fluent), Hindi (Fluent), Bengali (Native)

Hobbies: Sports (Badminton, Tennis, Padel), Art (Sketching, Painting, Digital Art), Backpacking, Photography, Cooking, Reading